ABSTRACT

As a renewable, biodegradable, and non-toxic fuel in recent years, biodiesel has garnered considerable interest from researchers, governments, and businesses. However, due to their extraordinarily high cost from being used primarily as food resources, a number of feedstocks have been shown to be impractical or unfeasible. Despite its disadvantages, such as its high free fatty acid and water levels, used cooking oil is seen to be the most promising biodiesel feedstock. The pre-treatment and use of waste cooking oil (WCO) for the synthesis of biodiesel utilising various techniques, multiple reactor types, various types and quantities of alcohol, and catalysts are all well covered in this review study.

A methanol-ethanol combination will combine the benefits of both alcohols in the transesterification process, which is the most widely used method for producing biodiesel. The purification and analysis of the generated biodiesel, operational factors that have a significant impact on biodiesel output, and a number of economic studies are also highlighted in this study. According to this analysis, WCO is a potential feedstock for the generation of biodiesel.

The goal of the current study was to examine various aspects of making biodiesel from neem oil and look into its fuel characteristics.Neem seeds have a 30–40% oil content. This study focuses on the production of mono alkyl esters, which are used to make biodiesel from neem oil. ‘Transesterification' is the process used. At various temperatures and with various molar ratios of neem oil and methanol, the ideal conditions to produce the highest production of biodiesel were explored. The most likely outcome, which included a layer of glycerine and soap, was examined using a 3:1 molar ratio of methanol and neem oil at temperatures ranging from 55 to 61°C. It was clear that the calorific value, density, and kinematic viscosity of biodiesel fuel all fell within acceptable ranges. The ASTM 6751-02 suggested range for kinematic viscosity for biodiesel is 1.90 to 6.0 centistokes. If biodiesel replaces just a small portion of petroleum and has better fuel qualities with reduced emissions of unburned hydrocarbons and carbon monoxide but greater levels of oxides of nitrogen, it could play a significant role in the future.

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CHAPTER 1: INTRODUCTION

One of the most appealing alternative fuels being produced right now is biodiesel made from renewable resources, which has minimal emissions and desirable chemical properties including being non-toxic, biodegradable, and carbon neutral. Additionally, unlike non-renewable fuel, biodiesel fuel may be used in any normal diesel engine (petroleum or diesel). Non-renewable fuel also has a short lifespan and releases pollutants including lead, hydrocarbons, nitrogen, sulphur, and carbon oxides. The sources of cooking oil vary across the world. Plant-based lipids like maize oil, margarine, coconut oil, palm oil, olive oil, soybean oil, grape seed oil, and canola oil or animal-based lipids like butter, ghee, and fish oil serve as their primary building blocks. Due to its low cost compared to other sources like coconut, maize, or soybean plants, oil palm is the most popular source of cooking oil in India. These lipids are trans-esterified to create biodiesel. Previous research has demonstrated that a variety of vegetable oils, including soybean, palm, and sunflower oil, may be used to make biodiesel.

A tree belonging to the militance family by the name of neem is found across Asia. The botanical name for it is Azadirachta indica. The huge, evergreen tree has a diameter of up to 1.8 to 2.4 metres and a height of 12 to 18 metres. 40% of the seeds are oil, which has great potential for making biodiesel. Its molecular weight, viscosity, density, and flash point are all higher than those of diesel fuel. Neem oil often has a light to dark brown colour, is bitter, and smells strongly like a mixture of garlic and peanuts.

SCOPE The scope of this project is to produce a good quality Biodiesel from mineral oils and is free from sulfur content, more ecofriendly fuel. Biodiesel produced from waste cooking oil and neem oil offers high cetane number, lower ash percentage and lower carbon residue. Also Improve engine ignition and reduced exhaust toxic emissions, and overall performance of the diesel engine.

OBJECTIVES

- To produce biodiesel from waste cooking oil and neem oil by using transesterification process.
- Conduct viscosity, flash and fire point, calorific value and emission test of produced biodiesel.
- Compare the obtained value with diesel and analyzing the performance of engine.

1.1 BIODIESEL

The American Society for Testing and Materials (ASTM) defines "biodiesel" as a fatty acid mono alkyl ester or fatty acid methyl ester produced from renewable feed supplies, such as vegetable oils. Contrary to regular diesel, the name "bio" denotes the biological source of biodiesel. The colour of biodiesel ranges from light to dark yellow and it is transparent. It has a boiling point exceeding 200 C, a flash point of 145–175 C, a distillation range of 195–325 C, and a vapour pressure of less than 5 (mm Hg at 22 1 C). It also possesses steady reactivity, a faint musty/soapy odour, is biodegradable, and is insoluble in water.

Biodiesel has a calorific value of roughly 37.27 MJ/kg. This is 9% less petroleum diesel than usual Number 2. More so than the manufacturing technique, variations in biodiesel energy density rely on the feedstock utilised. These variances, however, are lower than for petroleum diesel. It's been asserted that biodiesel has superior lubricity and more thorough combustion, which increases engine output and somewhat offsets petroleum diesel's higher energy density. Although biodiesel lacks the sulphur components that give petroleum diesel much of its lubricity, it nevertheless has promising lubricating characteristics and higher cetane ratings than low sulphur diesel fuels, and it frequently acts as an addition to ultra-low-sulphur diesel fuel to help with lubrication.

Biodiesel Higher lubricity fuels may lengthen the useable life of high-pressure fuel injection machinery that depends on the lubrication provided by the fuel. This might contain high pressure injection pumps, pump injectors, and fuel injectors, depending on the engine.

In the retail diesel fuel industry, blends of conventional hydrocarbon-based diesel and biodiesel are most often supplied. A large portion of the world utilises the "B" factor to indicate the proportion of biodiesel in any gasoline mixture.

Figure 1.1 biodiesel [1]

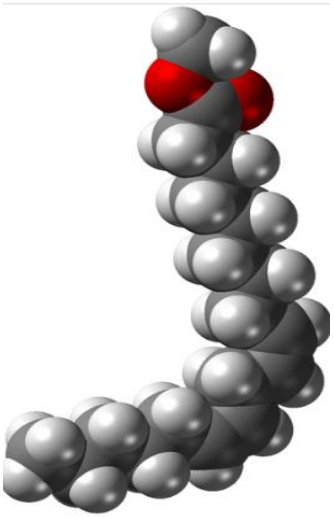


Figure 1.2 structure of biodiesel [2]

- B100 is the name for 100% biodiesel.
- B20 fuel contains 80% petroleum and 20% biodiesel.
- The label reads "7% biodiesel, 93% petroleum diesel." B7”
- B5 is marked as 5% biodiesel and 95% petroleum fuel.
- B2 The label reads: 2% biodiesel, 98% petroleum diesel.

Without major changes, blends of 20% biodiesel and below can be used in diesel equipment however, some manufacturers do not extend warranty coverage if equipment is harmed by these blends. The ASTM D7467 standard covers the B6 through B20 mixes. In order to prevent maintenance and performance issues, biodiesel can also be utilised in its pure form (B100), however this may need certain engine modifications. You may blend B100 with petroleum fuel by doing the following:

- mixing in tanks before being delivered to a tanker truck at the production location
- mixing splashes in the tanker truck (adding specific percentages of biodiesel and petroleum diesel)
- Two components arrive to the tanker truck simultaneously during in-line mixing.
- Biodiesel, petroleum diesel, and metered pump mixing metres are all set to X total volume.

1.2 APPLICATIONS

- In automobiles
- Air crafts
- Locomotives
- Generators
- Cleaning of oil spills
- Heating oil

1.3 WASTE COOKING OIL

Cooking oil residue from frying foods made of plant- or animal-based fats. A glycerol ester made up of many essential fatty acids, cooking oil can only be dissolved in organic solvents. Consuming cooking oil waste is bad for the environment and the health of consumers. The toxic compounds produced when the oil oxidises are thought to cause cancer when fried foods are consumed. WCO is commonly accessible but frequently discarded after use, especially after frying.

The fact that most used cooking oil from homes is dumped in the drainage and soil in India, where an estimated 540 000 tonnes of WCO from vegetable and animal fats are thrown each year without being handled as wastes, supports this claim. Therefore, using the WCO as the raw materials for more affordable, environmentally friendly biodiesel is one suggestion to address these problems. Due to the increase in WCO production, it may potentially be used as a feedstock for the manufacturing of biodiesel.

The characteristics of WCO are different from fresh oil because to chemical events such as oxidation, hydrolysis, polymerization, and material transfer between the food and the oil that take place during the frying process. In terms of density, kinematic viscosity, acid value, and iodine value, WCO has equivalent attributes to create biodiesel and fulfil the quality criteria. The amount of free fatty acids (FFA) in WCO is larger than in vegetable oil, which is important when choosing a catalyst to employ in the manufacturing process. Large-scale biodiesel production is constrained in environments with high FFA concentrations, but soap manufacture is encouraged. According to research, WCO has very

high values for kinematic viscosity, saponification, flash point, moisture content, and free fatty acids. WCO is 2 to 3 times less expensive than fresh vegetable oil, which significantly lowers overall processing costs. Since the cost of raw materials accounts for up to 75–90% of the total cost of producing biodiesel, it has a considerable impact on the retail price of the fuel. The usage of WCO as raw materials is independent of any crises or problems, such as the argument over whether to utilise food or fuel.

Figure 1.3 waste cooking oil.

1.4 NEEM OIL

Neem oil, often referred to as margosa oil, is a type of vegetable oil that is extracted from the fruits and seeds of the neem tree (Azadirachta indica), which is native to the Indian subcontinent but has since been spread to many other tropical regions. It is the most significant neem product now on the market and is utilised in both organic farming and medicine. Its molecular weight, viscosity, density, and flash point are all higher than those of diesel fuel. Neem oil often has a light to dark brown colour.

Figure 1.4 Neem oil

CHAPTER 2: LITERATURE REVIEW

Biodiesel has become a key source as a substitution fuel and making its place as a key future renewable energy source. As an alternative fuel for diesel engines, it is becoming increasingly important due to diminishing petroleum reserves and the environmental consequences of exhaust gases from petroleum fuelled engines.

2.1. GENERAL HISTORY OF BIODIESEL

Biofuels were once our primary fuel. The use of vegetable oils as an alternative fuel has been around for 100 years when the inventor of the diesel engine Rudolph Diesel first tested peanut oil, in his compression diesel engine. The first biodiesel manufacturing plant specifically designed to produce fuel was started in 1985 at an agricultural college in Austria. The first public demonstration of vegetable oil based diesel fuel was at 1900 World's Fair at France. Later, due to the widespread availability and low cost of petroleum diesel fuel, biodiesel gained little attention. But after the World War 2 and the oil crisis in 1970s, the importance of biodiesel increased. Over the last few years, there has been increasing amounts of research and interest in the production of biodiesel.

2.2. PREVIOUS WORKS

In 2007, Smruthi Rekha Dash of NIT Rourkela carried out studies on synthesis of biodiesel from vegetable oils by the method of alkali transesterification. After the work, it has been proposed virgin palm oil as one of the best alternatives for biodiesel synthesis. Vegetable oils are widely available from various sources, and the glycerides present in the oils can be considered as a viable alternative for diesel fuel. Biodiesel, which is synthesized from bio-oil, is a realistic alternative of diesel fuel because it provides a fuel from renewable resources and has lower emissions than petroleum diesel. It is biodegradable and contributes a minimal amount of net greenhouse gases or sulphur to the atmosphere. More specifically, biodiesel cuts down on the amount of carbon dioxide, hydrocarbons, and particulate matter released into the environment

In 2009, Piyanuch Nakpong and Sasiwimol Woothikanokkhan investigated the production of biodiesel from three mixtures of vegetable oil and used cooking oil by alkali transesterification. The effect of used cooking oil content in feedstock including jatropha, roselle, coconut oils were tested and optimised.

. In 2011, Prafulla D Patil, Veer Gnaneswar and Harvind L Reddy Of New Mexico State University at USA carried out a comparative study between acid transesterification (using sulphuric acid) and microwave irradiation process for biodiesel production. The studies concluded that the latter one is the efficient method.

In 2012, Dharunde Dhiraj S and Prof. Deshmukh Mangesh of gcoe at Amaravathi analyzed the biodiesel production from animal fat and its impact on diesel engine. They used chicken fat, mutton fat and beef tallow as feedstocks. The results showed that the biodiesel produced from animal fat can be used in diesel engines with turbocharging.

In 2015, Ishanullah, Sumaira shah and Muhammed Ayaz of king Fahd university of petroleum and minerals at Saudi Arabia carried out studies on synthesis of biodiesel from second generation oils. They extracted oil from spirogyra, an algael species using n hexane and diethyl ether. Further biodiesel has been produced from that oil by conventional transesterification method.

In 2019 Z. Yaakob, M. Mohammad Overview of biodiesel production from waste cooking oil Pretreatment and usage of WCO for Biodiesel production, Biodiesel is defined as the monoalkyl esters of long chain fatty acids derived from renewable feed stocks like vegetable oils or animal fats. While most of the properties of biodiesel are comparable to petroleum based diesel fuel, improvement of its low temperature flow characteristic still remains one of the major challenges while 7 using biodiesel as an alternative fuel for diesel engines. The biodiesel fuels derived from fats or oils with significant amounts of saturated fatty compounds will display higher cloud points and pour points they found a quality improvement and better fuel economy was obtained.

In 2022 M.U.H. Suzihaque , Habsah Alwi , Ummi Kalthum Ibrahim Biodiesel production from waste cooking oil, they found that an improvement in engine performance and emission. Also obtained good yield of biodiesel and reduced suffer and carbon residue. Using transesterification process.

CHAPTER 3: MATERIALS AND METHODOLOGY

The materials used for biodiesel production are waste cooking oil, neem oil, methanol and KOH.

In the process of making biodiesel, one ester combines with alcohol to create another ester and another alcohol. The vegetable oil (Neem oil) that contains triglycerides in this case is the ester. In a diesel engine, there are three different ways to employ neat vegetable oils.

- Pyrolysis.
- Micro emulsification.
- Transesterification

3.1 PYROLYSIS

Plant biomass may be transformed using a variety of technologies and processes into more valuable energy products. Biomass is heated to temperatures between 400 and 500 degrees Celsius during pyrolysis, a thermochemical process that yields bio-oil that mimics crude oil. Bio-oil undergoes hydrotreating and hydrocracking to be converted into useful transportation fuels. In hydrotreating, sulphur and oxygen are removed from the bio-oil by a reaction between hydrogen and the oil. In hydrocracking, the hydrogen-reacted hydrotreated bio-oil is reacted once again to produce shorter chains of hydrocarbons that adhere to the requirements of gasoline and diesel fuels.

3.2 MICRO EMULSIFICATION

Micro-emulsion, as defined by IUPAC, is a dispersion comprised of water, oil, and surfactant(s) that is an isotropic and thermodynamically stable system with dispersed domain diameter varied roughly from 1 to 100 nm, often 10 to 50 nm.

Diesel fuel, vegetable oil, alcohol, and surfactant and cetane improver in the proper ratios are the ingredients of a biodiesel micro-emulsion. Alkyl nitrates are utilised as cetane improvers, higher alcohols are used as surfactants, and alcohols like methanol and ethanol are used to reduce viscosity. Vegetable oils, an ester, a dispersant (co-solvent), alcohol, a surfactant, and a cetane improver can all be used to create a micro-emulsion, with or

without the addition of diesel fuels. The maximum viscosity criterion for diesel fuel was reached by all micro-emulsions including butanol, hexanol, and octanol. In the micellar solubilization of methanol in triolein and soybean oil, 2-octanol is an efficient amphiphile.

Through the explosive vaporisation of the low boiling components within the micelles, micro-emulsions can enhance spray characteristics. The biodiesel's viscosity is reduced via micro-emulsion, and its cetane number and spray characteristics are improved. The short-term performances of both ionic and non-ionic micro-emulsions of aqueous ethanol in soybean oil were almost satisfactory.

3.3 TRANSESTERIFICATION PROCESS

The most well-liked and effective approach for using clean vegetable oils is transesterification. Researchers E. Duffy and J. Patrick started it in 1853, several years before the first working diesel engine was created. This method was created in the 1940s to enhance the separation of glycerine for the purpose of making soap.

For the esterification process, an acid catalyst was utilised, and for the transesterification reaction, an alkaline catalyst (KOH or NaOH) was used. Vegetable oil must be trans esterified in order to create methyl esters, which takes raw oil, 15% methanol, and 5% sodium hydroxide on a mass basis. In the presence of a catalyst (KOH or NaOH), one mole of glyceride interacts with three moles of methanol to form methyl esters.

The temperature of the reaction must be between 55 and 65 degrees Celsius. Most of the time, the temperature is kept below methanol's typical boiling point (65°C), negating the necessity for pressurising the reactor.

First, you need to get to an equilibrium temperature range of 55 to 65 °C. Second, the product mixture was constantly agitated for glycerine separation before being allowed to settle naturally in a separating funnel. After gravity has settled after 24 hours, two separate strata form. Glycerol made up the lowest layer, whereas ester made up the top layer. The bottom layer was removed. The isolated ester often contains some soap and 3% to 6% methanol. The methanol can be extracted by vaporisation if the soap concentration is low enough (between 300 and 500 ppm), and this methanol is typically dry enough to be recycled back into the process.

The biodiesel must be washed to remove any remaining free glycerine, methanol, soap, or catalyst after the methanol has been removed. Before making an emulsion of the pollutants themselves, it is commonly done. In order to eliminate the catalyst found in ester, biodiesel was combined with some warm water (about 10% volume) and let to settle naturally for an additional 24 hours. The water is separated and the moisture is removed before the catalyst dissolves. Until the wash water is no longer picking up soap, the washing operation is often repeated several times.

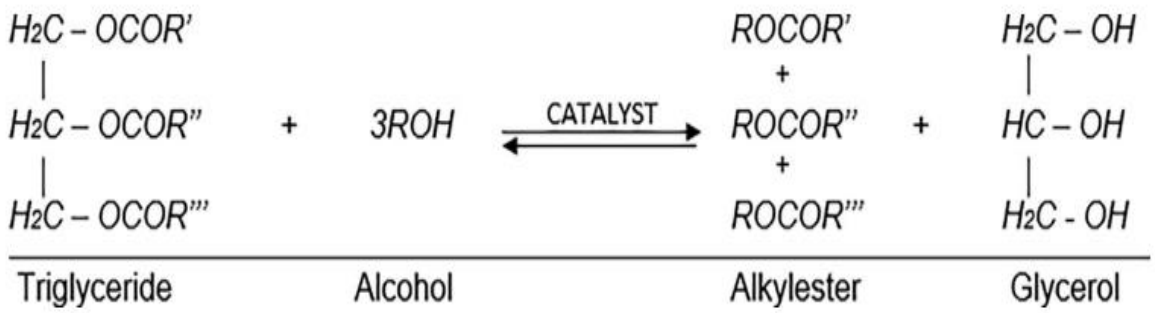


Figure 3.1 Transesterification reaction.

The total amount of water will normally be one or two times the volume flow rate of the biodiesel, even if the grey water from later washes can be utilised as the supply water for the earlier wash processes. I heated the biodiesel mixture to a temperature above 100°C and added 10% water to it. Both the remaining methanol and the water evaporated. Acids can be used to decrease the water content. The catalyst will be neutralised by a weaker organic acid, such citric acid, and a soluble salt will result.

Because the salts are simpler to remove than the soap, stronger inorganic acids like hydrochloric, sulfuric, or phosphoric can be employed to break the soap, lowering the water need to 5% to 10% of the biodiesel. After biodiesel has settled out, any remaining water droplets can be removed using a flash evaporator. Although I utilised a water washing approach to separate the remaining methyl esters, you may instead use solid absorbents like magnesium silicate in place of water washing. Soap, catalysts, and free glycerol are among the pollutants that are absorbed by the fine powder of absorbents.

Glycerol contains soaps, catalyst, methanol, and other contaminants when esters and glycerol are separated. Purity of the glycerol is therefore required to improve the current condition. Glycerine was used with strong hydrochloric acid (HCl) to break the soap and neutralise the catalyst.

Soap + Hydrochloric Acid → **Fatty Acid + Salt**

The free fatty acids in glycerine may be easily separated using a centrifuge since they are not soluble in the substance. Vaporization can be used to get rid of methanol. The last impurity is salt. The separation of methyl esters and glycerine during the biodiesel synthesis process may be hampered by large levels of free fatty acids (FFA), which is of particular concern**.**

Fatty Acid +NaOH → **Soap + Water**

Fatty Acid + Methanol **H2SO4** → **Methyl Ester + Water**

The transesterification process is slowed by acid-based catalysation. At a moderate temperature (50 to 61°C), the two-step method of base- and acid-catalysed transesterification produced a full reaction. The discovered methyl ester was ultimately processed to create the biodiesel that we required. (Rakibul, 2013).

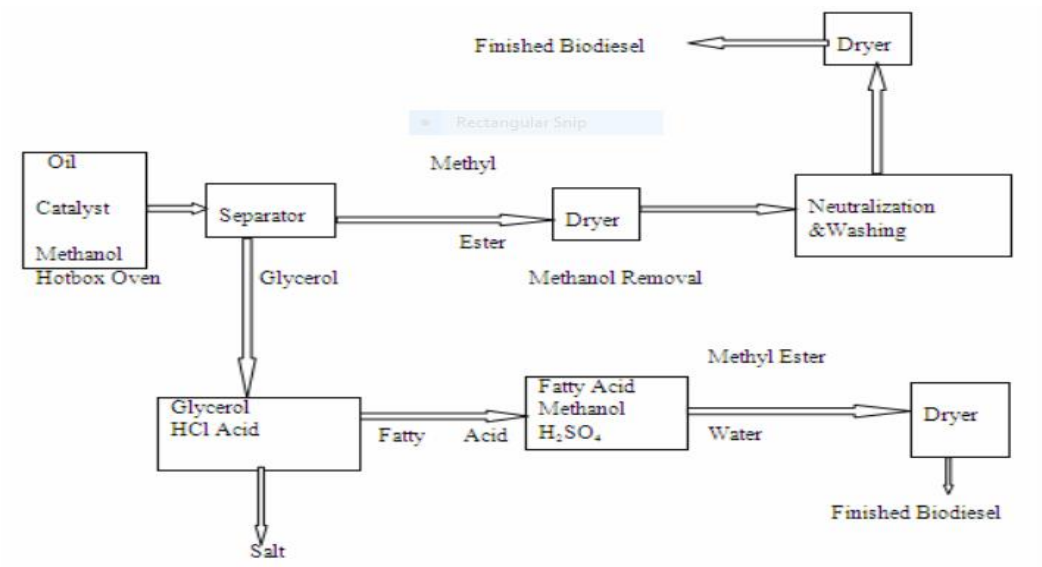


Figure 3.2 schematic representation of biodiesel processing

3.4 EQUIPMENTS USED

3.4a REEDWOOD VISCOMETER

A viscometer, sometimes known as a viscosimeter, is a device used to gauge a fluid's viscosity. A device known as a rheometer is used for liquids whose viscosities change depending on the flow circumstances. As a result, a rheometer may be thought of as a particular kind of viscometer. Viscometers can only measure only one type of flow. Typically, an item passes through a fluid while it is motionless, or the object is immobile while the fluid is passing by it. The viscosity of a fluid is determined by the drag it produces when a surface moves relative to it. For there to be laminar flow, the flow conditions must have a sufficiently low Reynolds number.

Figure 3.3 REDWOOD VISCOMETER

3.4b BOMB CALORIMETER

To calculate the calorific values of solid and liquid fuels, one uses a bomb calorimeter. It is made up of a powerful steel shell called a bomb.

It consists of a base that is fastened to the bomb's body and supports the platinum crucible. An oxygen supply connector and a product release valve are located on the top of the bomb.

For the test, one gramme of powdered sample coal is used, and 2000 cm3 of water is added to the calorimeter. The platinum crucible is filled with the sample. The bottom end of the two electrodes is attached to the iron fuse wire that encircles the sample of coal.
The fuse wire is connected to an electrical circuit via the electrodes, which protrude through the bomb's base. By cutting the electric circuit, the coal may be ignited.

The explosive is put inside a copper container filled with water. The water within the calorimeter is stirred by a stirring mechanism. Another container that serves as a heat insulator is used to house the calorimeter that contains the bomb. A thermometer is used to gauge the calorimeter's water's temperature. When the pressure gauge in the oxygen cylinder reads 25 atmospheres, oxygen is introduced to the bomb through the value and the oxygen cylinder is connected to the bomb. By running a current via the fuse wire, the gasoline is lit. Following the maximum temperature being attained, the temperatures of both begin to rise, and the thermometer's readings are taken at intervals of one minute for 10 minutes. The temperature then gradually begins to drop. For a further five minutes, measurements are collected at regular intervals when the temperature decline indicates a consistent rate.

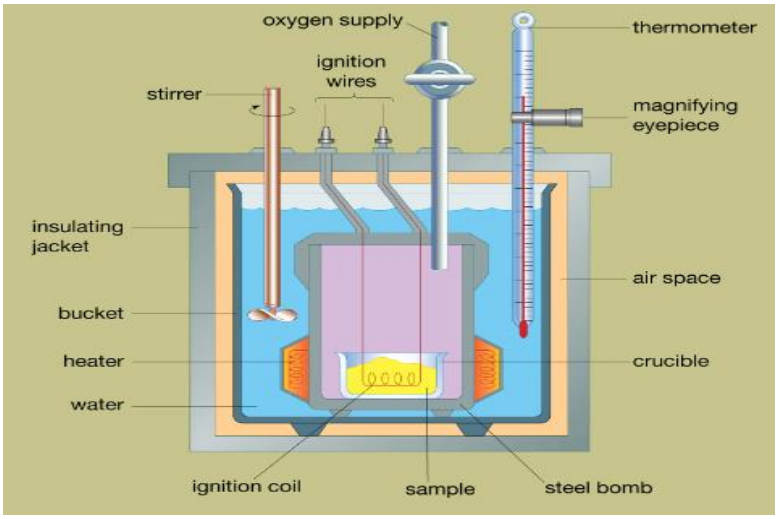


Figure 3.4 schematic representation of bomb calorimeter [2]

3.4c FLASH AND FIRE POINT APPARATUS

FLASH POINT The lowest liquid temperature at which, under specific specified circumstances, a liquid emits vapours in sufficient quantity to be capable of generating an ignitable vapour/air combination is known as a material's flash point.

Figure 3.5 Flash and fire apparatus

FIRE POINT The lowest temperature at which the vapour of a fuel will continue to burn for at least five seconds after being ignited by an open flame of standard dimension is known as the fire point of that fuel. A material will momentarily burn at the flash point, a lower temperature, but it's possible that not enough vapour will be created to keep the fire going. Only material flash points are often listed in tables of material characteristics. Although this is not a replacement for testing to determine whether the fire point is safety essential, it may generally be considered that the fire point is around 10 °C higher than the flash point.

3.4d EXHAUST GAS ANALYZER

An exhaust gas analyser, often known as an exhaust carbon monoxide (CO) analyser, is a device used to monitor carbon monoxide and other gases released into the atmosphere as a result of improper combustion. Infrared gas sensors and chemical gas sensors serve as the foundation for CO sensors (as well as other types of gas sensors). Carbon monoxide sensors are used to gauge the test's CO level.

Figure 3.6 exhaust gas analyser

CHAPTER 4: PREPARATION OF BIODIESEL

4.1 PREPARATION OF BIODIESEL FROM WASTE COOKING OIL

Waste cooking oil (200 ml) is collected and cooked in a beaker it is heated to 60 degrees Celsius to make biodiesel. 50 ml of methanol and 5 grammes of KOH are poured to a conical flask, agitated with a magnetic stirrer, and then allowed to thoroughly combine. After properly combining methanol and KOH solution, oil that has already been heated is added. This process, called transesterification, is then allowed to settle for a day.

Glycerol is the bottom product, while the top product is biodiesel. The biodiesel is separated, then washed and dried. We get a yield of 90%, and it is prepared for testing.

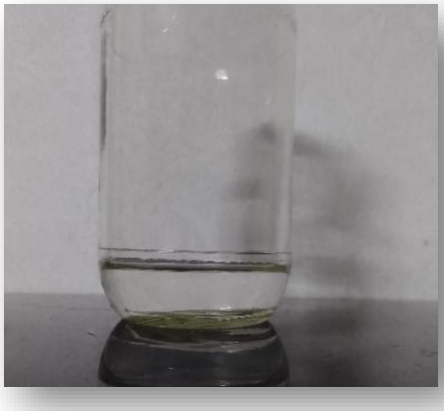


Figure 4.1 Methoxide solution

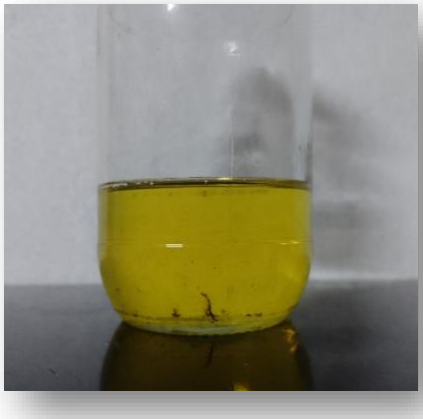


Figure 4.2 waste cooking oil

Figure 4.3 biodiesel & glycerol

Figure 4.4 neem oil biodiesel & glycerol

Figure 4.5 Neem oil

CHAPTER 5: RESULTS AND DISCUSSION

5.1 KINEMATIC VISCOSITY

RESULTS OBTAINED

Viscosity of WCO biodiesel

TEMP	VISCOSITY (c st)
30	6.13
40	4.86
60	3.28
80	2.37
100	1.83

. Table 5.1 kinematic viscosity of WCO biodiesel

The kinematic viscosity of neem oil biodiesel

TEMP	VISCOSITY c st
30	21.38
40	18.36
60	17.5
80	15.75
100	13.89

Table 5.2 kinematic viscosity of neem oil biodiesel

5.2 FLASH AND FIRE POINT

FLASH & FIRE POINT OF WCO BIODIESEL

TEMP ^{0}C	FLASH POINT	FIRE POINT
50	NO FLASH	NO FIRE
55	NO FLASH	NO FIRE
60	NO FLASH	NO FIRE
65	FLASH	FLASH
70	FLASH	FLASH
75	FIRE	FIRE

Table 5.3 flash & fire point of WCO biodiesel

FLASH AND FIRE POINT OF NEEM OIL BIODIESEL

The flash point of neem oil bio diesel is 147 ^{0}c

The fire point of neem oil bio diesel is 160 ^{0}c

5.3 CALORIFIC VALUE

The calorific value of waste cooking oil biodiesel is = 37.18 MJ/kg

The calorific value of neem oil biodiesel is = 28.23 MJ/kg

5.4 MEASUREMENT OF EXHAUST EMISSIONS

EXHAUST EMISSIONS FROM WCO BIODIESEL

	O2 (%)	CO (ppm)	CO2 (%)	NO (ppm)	NOX (ppm)
Biodiesel (WCO)	16.2	820	1.0	53	82

Table 5.4 exhaust emissions from WCO biodiesel.

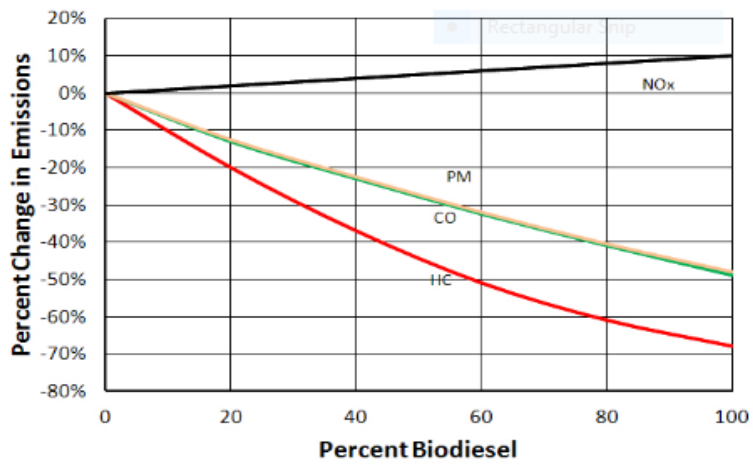
5.5 COMPARISON BETWEEN DIESEL AND BIODIESEL

In comparison to diesel, biodiesel produces 11% less carbon monoxide, 10% less particulate matter, and 78% less net carbon dioxide.

Diesel engines are more efficient than gasoline engines by 20–40% and provide higher torque at lower rpms. Vehicles may operate on biodiesel without having any changes, which further reduces pollutants when used in diesel engines.

O_2 (%)	CO (ppm)	CO_2 (%)	NO (ppm)	NO_X (ppm)
16.	928	1.9	38	102
16.3	971	1.1	55	109
16.6	729	2.3	46	104
16.7	890	1.4	33	89
17.2	551	0	49	100
18.7	583	0	50	95
17	745	1.1	55	100

Table 5.5 emissions from diesel engine



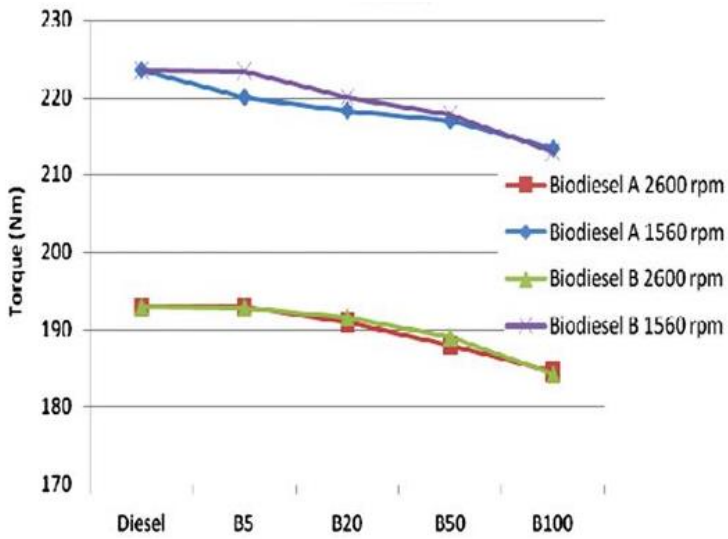


Figure 5.1 & 5.2 graph showing emission and torque characteristics of biodiesel

The way filling stations handle biodiesel in comparison to conventional fuels is the second significant issue with the fuel. It must be kept at the proper temperature for storage because it is manufactured from ingredients with a vegetable foundation. It can develop mould if kept in a heated storage tank for an extended period of time. On the other hand, if it is kept at too-cold temperatures, it can thicken and be challenging to dispense. Concerning issues are those brought on by enhanced lubricity while switching from conventional systems to biodiesel. High lubricity aids in delaying the onset of fuel system wear and tear, but it can also cause fuel filter blockages by releasing deposits on tank walls and pipes.

CHAPTER 6: CONCLUSION

The following important aspects of biodiesel production from waste cooking oil (WCO) were addressed. The most effective fossil fuel substitute has been found to be biodiesel.

It performs better than petroleum-based fuels since it is renewable, biodegradable, and non-toxic. WCO has a lot of promise as a source of biodiesel given how much waste cooking oil is produced globally. Due to its cheaper cost than other raw materials, WCO is more likely to have a bigger role in biodiesel production in the future. Contrarily, WCO requires a variety of pre-treatment processes to get rid of solid impurities, cut down on FFA, and lower water content.

The temperature, reaction duration, catalyst concentration, and the oil to alcohol molar ratio are only a few of the variables that affect the transesterification process, which is utilised to create biodiesel. In comparison to diesel, biodiesel provides a variety of benefits, including a greater cetane rating, a lower ash content, and a smaller carbon residue. Additionally, the engine ignition and exhaust performance have been enhanced. Neem oil has the potential to be a useful alternative fuel. Oil cannot, however, eliminate dependence on foreign oil in any reasonable amount of time. Increasing Neem oil production would need a significant investment of resources. The most recent developments in instrumentation technology are addressing these demands.

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